

Cranberries vs antibiotics to prevent urinary tract infections: a randomized double-blind noninferiority trial in premenopausal women.



BACKGROUND: The increasing prevalence of uropathogens resistant to antimicrobial agents has stimulated interest in cranberries to prevent recurrent urinary tract infections (UTIs). **METHODS:** In a double-blind, double-dummy noninferiority trial, 221 premenopausal women with recurrent UTIs were randomized to 12-month prophylaxis use of trimethoprim-sulfamethoxazole (TMP-SMX), 480 mg once daily, or cranberry capsules, 500 mg twice daily. Primary end points were the mean number of symptomatic UTIs over 12 months, the proportion of patients with at least 1 symptomatic UTI, the median time to first UTI, and development of antibiotic resistance in indigenous *Escherichia coli*. **RESULTS:** After 12 months, the mean number of patients with at least 1 symptomatic UTI was higher in the cranberry than in the TMP-SMX group (4.0 vs 1.8; $P = .02$), and the proportion of patients with at least 1 symptomatic UTI was higher in the cranberry than in the TMP-SMX group (78.2% vs 71.1%). Median time to the first symptomatic UTI was 4 months for the cranberry and 8 months for the TMP-SMX group. After 1 month, in the cranberry group, 23.7% of fecal and 28.1% of asymptomatic bacteriuria *E coli* isolates were TMP-SMX resistant, whereas in the TMP-SMX group, 86.3% of fecal and 90.5% of asymptomatic bacteriuria *E coli* isolates were TMP-SMX resistant. Similarly, we found increased resistance rates for trimethoprim, amoxicillin, and ciprofloxacin in these *E coli* isolates after 1 month in the TMP-SMX group. After discontinuation of TMP-SMX, resistance reached baseline levels after 3 months. Antibiotic resistance did not increase in the cranberry group. Cranberries and TMP-SMX were equally well tolerated. **CONCLUSION:** In premenopausal women, TMP-SMX, 480 mg once daily, is more effective than cranberry capsules, 500 mg twice daily, to prevent recurrent UTIs, at the expense of emerging antibiotic resistance. **TRIAL REGISTRATION:** isrctn.org Identifier: ISRCTN50717094.